

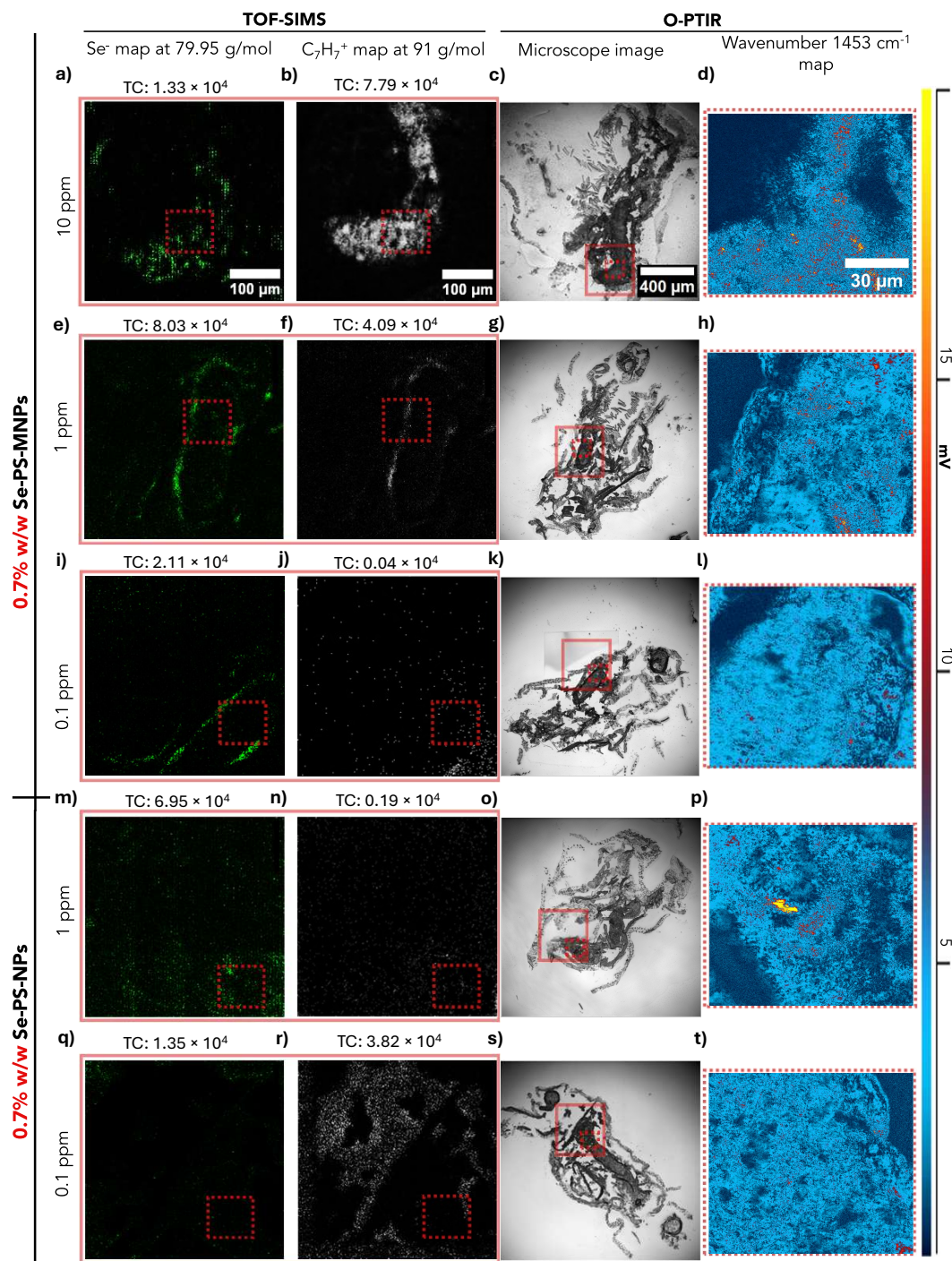


Figure 4. Localization of Se-PS-MNPs and Se-PS-NPs in histological slices of *Daphnia magna* exposed to 0.1, 1 and 10 ppm Se-PS-MNPs and 0.1 and 10 ppm Se-PS-NPs at 0.7% w/w Se-to-polymer ratio using nano-TOF-SIMS with negative ion selenium mapping (green) at 79.95 g/mol and positive ion styrene ($C_7H_7^+$) mapping (blue) at 91 g/mol for a $400 \times 400 \mu m$ region located as a red square on the O-PTIR $10\times$ microscope image, and O-PTIR mapping at 1453 cm^{-1} with O-PTIR colour intensity scale shown on the right side. Red and yellow indicate stronger presence for 1453 cm^{-1} . Red dashed square is the location of the mapping indicated on the microscope image and nano-TOF-SIMS images;